

MTH 207 Exam 1 (Dr. Kim, Fall 2026)

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Rules

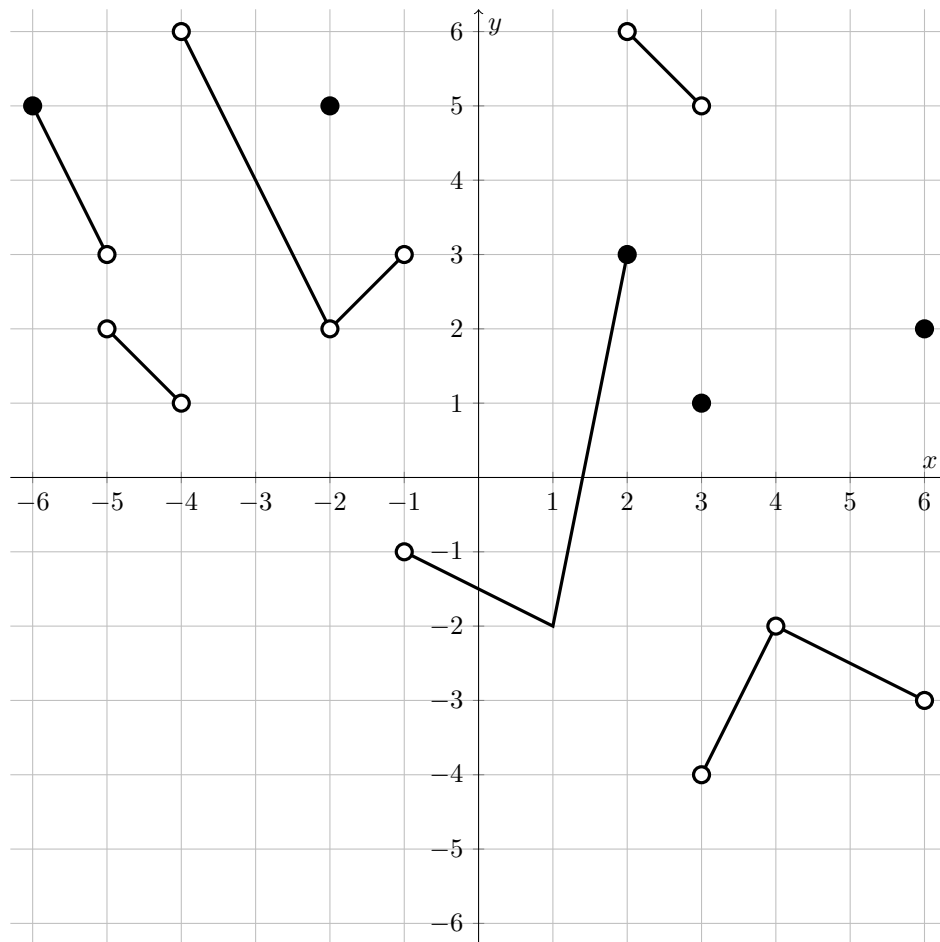
- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating “does not exist” without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

The exam length is 55 minutes. There are 100 possible points.

1. The graph below is $y = f(x)$. If a requested value does not exist, explain why.



(3pts) (a) Find $\lim_{x \rightarrow -2} f(x)$

(3pts) (b) Find $f(3) + f(6)$

(3pts) (c) Find $\lim_{x \rightarrow 4} f(x)$

(3pts) (d) Find $\lim_{x \rightarrow -4} f(x)$

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(8pts) 2. Given $f(x) = x^2$ find $f'(10)$ directly by a limit-based definition. (No credit will be given for applying derivative rules such as the Power Rule.)

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(8pts) 3. Find m and c so that

$$f(x) = \begin{cases} 2x - m & \text{if } x < 6 \\ x - c & \text{if } x = 6 \\ 10 - x & \text{if } x > 6 \end{cases}$$

is continuous at $x = 6$, applying the definition. 37

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(8pts) 7. Find $\lim_{x \rightarrow 6} \frac{1}{x - 6}$

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(8pts) 8. Find $\lim_{x \rightarrow \frac{\pi}{6}} \sin(x)$

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(8pts) 9. Find $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$

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(8pts) 10. Find $\lim_{x \rightarrow 0} \frac{x}{|x|}$ 118

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(8pts) 11. Find $\lim_{x \rightarrow 0} x^6 \sin(\frac{1}{x})$ 129

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(8pts) 12. Find $\lim_{x \rightarrow \infty} \frac{4x^2 - 5}{8x^2 + 9}$ 140

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